



CS2308 Spring 2026

Foundations of Computer Science II



Announcement: Yay!! Welcome to CS2308

Instructor	Dr. Keshav Bhandari (Pronunciation: K - SUB)
Class Duration	01/20/2026 to 05/13/2026 meeting Mon & Wed every week
Office Hours	Mon & Wed 11:00AM-12:00 PM, COMAL 207H
Zoom Link	https://txstate.zoom.us/j/7643554723? PIN (2627)
Email Address	mailto:k_b459@txstate.edu

Grader	Sandhya Neupane
Email	mailto:sandhya@txstate.edu
Zoom Link	https://txstate.zoom.us/j/7643554723? pwd=pXks286C5boMqKBigB4a5g3l6bRXNe.1
Office Hours	Via Zoom: Tue & Thu 2:00pm-3:00pm, Email for meeting confirmation

Class (CRN / Section)	Meeting Time	Location	Final Exam (click - Academic Cal.)
36702 / 256	09:30AM - 10:50 AM	Derrick Hall 00234	Wed. May 06, 2026 08:00-10:30 AM
33526 / 253	12:30 PM - 01:50 PM	Hines Academic Center 00201	Wed. May 13, 2026 11:00-01:30 PM

Welcome to the CS2308: Foundations of Computer Science II

The course is an introduction to Abstract Data Types (ADTs) including lists, stacks, and queues. Searching and sorting, pointers and dynamic memory allocation, and simple classes and objects also will be covered. The course is a continuation of [CS 1428](#). Prerequisite: [CS 1428](#) with a grade of "C" or better.



What's happening this week?



This only shows the tentative start date of the lecture series. We will meet twice a week Mon & Wednesday.

Assessments & Activities

Activity	Start Date	Description	Due Date	Status
Lecture 08: An OO Design	@April 20, 2026			Not started
Lecture 03: Algorithm Analysis	@February 18, 2026			Not started

 Activity	 Start Date	 Description	 Due Date	 Status
<u>Lecture 05:</u> <u>UNIX Prog</u>	@March 11, 2026			Not started
<u>Lecture 06:</u> <u>Abstract Data Structure</u>	@April 6, 2026			Not started
<u>Lecture 02:</u> <u>Pointers</u>	@February 9, 2026			Not started
<u>Lecture 07:</u> <u>Templates</u>	@March 23, 2026			Not started
<u>Lecture 01:</u> <u>Quick Recap</u>	@January 21, 2026	Explain how C++ programs are compiled into machine code and differentiate between compiled and interpreted languages. Use functions effectively in C++, including parameter passing, return types, scope, overloading, and lambda expressions. Organize multi-file C++ programs using proper file structure, header files, and variable scope. Work with arrays and STL containers (vectors, lists, stacks, maps) and select appropriate iteration strategies.		Not started

 Activity	 Start Date	 Description	 Due Date	 Status
		Build and run C++ projects using CMake and IDE-based workflows in CLion and Visual Studio Code.		
<u>Lecture 04: Struct Op Overloading and More</u>	@February 25, 2026			Not started
<u>Unit Labs 01</u>	@February 4, 2026			Not started
<u>Unit Labs 02</u>	@February 16, 2026			Not started
<u>Unit Labs 03</u>	@February 23, 2026			Not started
<u>Unit Labs 04</u>	@March 9, 2026			Not started
<u>Unit Labs 05</u>	@March 23, 2026			Not started
<u>Unit Labs 06</u>	@April 1, 2026			Not started
<u>Unit Labs 07</u>	@April 15, 2026			Not started
<u>Unit Labs 08</u>	@April 27, 2026			Not started
<u>Untitled</u>				Not

Aa Activity	📅 Start Date	≡ Description	📅 Due Date	⚙ Status
				starte d
<u>HW 01</u>	@January 26, 2026		@April 30, 2026	Not starte d
<u>HW 02</u>	@January 26, 2026		@April 30, 2026	Not starte d
<u>HW 03</u>	@January 26, 2026		@April 30, 2026	Not starte d
<u>HW 04</u>	@January 26, 2026		@April 30, 2026	Not starte d
<u>Quiz 01</u>	@March 11, 2026		@March 25, 2026	Not starte d
<u>Quiz 02</u>	@April 23, 2026		@April 30, 2026	Not starte d

Assessments & Activities

Aa Activity	📅 Start Date	⚙ Status
<u>Unit Labs 01</u>	@February 4, 2026	Not started
<u>Unit Labs 02</u>	@February 16, 2026	Not started
<u>Unit Labs 03</u>	@February 23, 2026	Not started
<u>Unit Labs 04</u>	@March 9, 2026	Not started
<u>Unit Labs 05</u>	@March 23, 2026	Not started
<u>Unit Labs 06</u>	@April 1, 2026	Not started

Aa Activity	📅 Start Date	⚙ Status
<u>Unit Labs 07</u>	@April 15, 2026	Not started
<u>Unit Labs 08</u>	@April 27, 2026	Not started

Books & Assessment Tools

▼ ZyBooks

Name	CS 2308: Foundations of Computer Science II
ISBN	979-8-203-14967-1
CODE	TXSTATECS2308BhandariSpring2026
Purpose	Assignment & Self Learning Interactive Book
Support	support@zybooks.com



How to join?

Click any ZyBooks assignment link from the Canvas Course Site. This will open a link in a new tab. If you are part of the BookSmart Program, get the access code from the BookStore (*They use <https://kortext.com/us/> to share the access code, check your email. You might have recieved **kortext** login detail*). If you are not part of the BookSmart Program, please subscribe directly.

✗ (Do not go to the ZyBooks website and create a new account with your personal email)

▼ TextBook (Optional and for reference)

Tony Gaddis, Starting out with C++: From Control Structures through Objects, 9th

Edition, ISBN: 0134498372 (8th edition is allowed!).

Tools

▼ CLion



Why?

We will use **CLion** in this course because it provides a low-friction, industry-aligned C++ development environment while remaining **free for students through JetBrains' academic license**. CLion comes with **CMake tightly integrated and pre-configured**, which aligns with how modern C++ projects are built and removes much of the manual setup that can distract from learning core C++ concepts. Its cross-platform consistency and strong debugging and code-navigation support allow us to focus on program design and language fundamentals rather than tooling issues. **Students who already know what they are doing may use VS Code**, provided they can correctly configure the compiler, CMake, and build tasks on their own.

Windows users are strongly advised to install Git Bash, as it provides a Unix-like terminal consistent with class instructions and common CMake/Git workflows. **Lightweight or legacy IDEs such as Code::Blocks are not recommended for this course**, as they do not align well with the CMake-based workflow used in modern C++ development.

1. [Click here to download CLion](#) (Select the right version)
 - a. Apply for CLion Student Pack [here](#) (no money required)
(Make sure to use `edu` email. You will get an email, click the link accept the agreement, now open CLion and activate the license by logging in your JetBrains account, if you forgot your password reset the password)
2. [Click here to download GitBash](#) (Only for windows users)
3. [Click her to download VSCode](#) (Only for experience users)

Penalties for lateness

Some sections of this course are **self-paced**, meaning there are **extended deadlines** for assignments, projects, and take-home quizzes. However, these extensions are not unlimited. **If you fail to submit your work by the extended deadlines, you will likely lose the opportunity to revise it.**

Although some sections may not have intermediate deadlines, the **final deadline is close to last day of the class**. Please do not interpret this flexibility as permission to procrastinate. Effective **time management is a critical professional skill**, and at this stage of your academic development, you are expected to plan your work responsibly and complete assignments on time.

Begin early, break the work into smaller milestones, and set **personal deadlines** well before the course ends. Last-minute submissions often result in unnecessary stress and reduced quality of work.

If you need guidance on creating a realistic timeline or setting your own deadlines, you are encouraged to **discuss your plan with me** during office hours or by appointment.

Rubrics

Grade Distribution

Attendance	05%	Via Canvas Attendance, must attend at least 80% to get full credit, otherwise you will earn proportional credit.
Knowledge Tests	05%	Via Canvas Quiz, after completion of each chapters.
Unit Labs	10%	Unit Labs are in-class labs after each chapter where students solve open-ended problems with guidance and collaboration. Code is submitted via a shared GitHub repo, and effort matters more than full

		solutions. AI use is allowed but must be documented in an <code>acknowledgement-of-ai-usage.md</code> file, and must contain full prompts and responses. Full credit requires attending 6 labs; fewer labs earn proportional credit. In the AI Usage document, submit markdown explaining how you used AI (or explicitly state no AI used) and paste the full prompts and AI responses exactly as generated by the system you used. Additional GitHub Requirements: Students must create a private GitHub repository named <code>CS2308SPRING2026LABS-NETID</code> (example: <code>CS2308SPRING2026LABS-K_B459</code>). Add keshavsbhandari (instructor) and sandhyaneupane (TA) as collaborators. The repository must follow this structure: <pre>CS2308SPRING2026LABS-K_B459 -- LAB01 -- files... -- LAB02 -- files... -- ...</pre>
4 Home Works	20% (05% x 4)	There are four homeworks (HW1–HW4), each accessed via Canvas and completed in ZyBooks. For each homework, you must submit a corresponding AI Usage assignment (e.g., <i>HW1: AI Usage</i>). These AI Usage assignments are not graded in Canvas but are reviewed to verify the credibility of your work. Timing is critical: if the AI Usage assignment is missing or not submitted immediately after the homework, the homework grade will be 0. In the AI Usage assignment, submit plain text explaining how you used AI (or explicitly state no AI used) and paste the full prompts and AI responses exactly as generated by the system you used.
Quiz-01 / Quiz 02	20% (10%	Quiz 01/02 (MCQs - Code analysis) consists of two required attempts: a take-home version (unproctored,

	x 2)	AI tools, and notes allowed) and a proctored version (no resources). Your final score will be calculated as a weighted average, where the take-home attempt counts less and the proctored attempt counts more to reflect its importance. For example, if the take-home is weighted at 40% and the proctored at 60%, your final score = $(0.4 \times \text{take-home score}) + (0.6 \times \text{proctored score})$. This ensures practice with support tools while giving greater credit to the supervised attempt. Curve - Only on Special Circumstances (will be explained in class).
Project	15%	Projects may be individual or in pairs (max 2 students). Teams must be formed early and reported to the grader. Submit the project via GitHub and Canvas. Include an <code>acknowledgement-of-ai-usage.md</code> file documenting any AI use. Additionally, record a ≤5-minute code walkthrough video, upload it to YouTube as unlisted, and submit the link. Students may be asked to demo or explain their project if needed.
Final Exam	25%	MCQs (15%): This will be in person, MCQs and focused more on Code Analysis like Quizzes. You will be given past questions to review. You will use scantron (bonus points for adding your name and ANumber accurately in scantron sheets) Open-Ended Questions (10% of Final): The final exam includes 6 open-ended questions, of which students may attempt any 4. These questions assess conceptual and critical understanding of code, not just code writing. Responses may include explanations, pseudocode, or actual code, but emphasis is on reasoning. Typical tasks include identifying bugs or logical flaws, analyzing inefficient or redundant code, recognizing critical regions, and spotting potential errors or edge cases.

Grading Policy

≥90	A
≥80	B
≥70	C
<70	F

AI Usage Policy (CS Coding Courses)

AI tools are **allowed and encouraged** when used **responsibly and for learning**. Students are expected to first engage with the **core concepts** and then use AI in an **exploratory and supportive role**, such as:

- clarifying concepts and resolving confusion (always fact-check - AI can hallucinate),
- breaking a problem into **smaller subproblems** and using AI interactively on those parts,
- studying **similar example problems** and writing your **own solution** based on understanding,
- diagnosing **bugs or logical errors** in your own code,
- supporting learning alongside **debuggers and other development tools** to understand code flow.

Using AI to generate **entire solution code** and submitting it as-is defeats the learning objective and is not acceptable.

Disclosure Requirement (Applies to Unit Labs, Homeworks, and Projects)

If you use AI, you **must submit an** **acknowledgement-of-ai-usage** **file/assignment**, clearly stating:

- **how** AI was used (or explicitly state *no AI used*),
- the **full prompts** you provided,
- the **full responses** generated by the AI (copy-paste).

As long as your submission demonstrates **genuine learning intent rather than solution-seeking**, you are fine.

Relying on AI as a shortcut creates **knowledge debt** - you may earn a grade, but lose competence. That path leads to being **unemployable**.

Use AI to **learn**, not to bypass thinking. Do not try to outsmart the process.

Course Policies

▼ Academic Integrity

All students are expected to maintain the highest standards of academic integrity. This includes, but is not limited to, avoiding actions such as cheating, plagiarizing, and providing false information. Any student found to be in violation of the academic integrity policy will face disciplinary action, which could include a failing grade for the course and further sanctions from the university.

▼ Attendance and Participation

Regular attendance and active participation in class are crucial for understanding the course material and doing well in the course. If you are unable to attend a class, please email the instructor in advance.

If you miss significant portion of the class you might not get points in attendance and participation.

▼ Electronics Policy

Use of electronics, such as laptops and cell phones, is allowed for note-taking and course-related purposes only. Any use of electronics that disrupts the class or distracts other students is not permitted.

▼ Office Hours

Office hours are a time for you to ask questions, discuss the course material, or seek help on assignments. If you cannot make the scheduled

office hours, please email the instructor or co-instructor to arrange an alternative time.

▼ **Communication**

All course-related communication will be done through email (Please clearly mention your name and class) or the discussion board on the course website. Please check your email and the course website regularly for updates.

▼ **Accommodations**

If you require any accommodations for a disability or other reasons, please contact the instructor as soon as possible to make the necessary arrangements.

Any student with needs requiring special accommodations should contact the office of disability services at the LBJ student center (5-5.1). Students who qualify for extra time for exams must take their test with ATSD and schedule it at the same time the test is given in class. Note: You must submit your request to ATSD at least 2 business days before the exam date!

▼ **Tutoring**

CS Lab tutors (good for help on Programming Assignments)

<https://cs.txstate.edu/resources/labs/tutoring/>

CLC tutoring (good for help on understanding the material)

<https://hlsamp.cose.txstate.edu/clc/clc-tutoring.html>

▼ Withdrawl Policy

You must follow the withdrawal and drop policy set up by the University

<http://www.registrar.txstate.edu/registration/dropping-or-withdrawing.html>

For rest of the deadlines regarding withdrawal please use university calendar

Academic Calendar

undefined

★ <https://www.registrar.txst.edu/registration/ac/academic-calendar.html>

